VoiceOps 2026 Capstone Report

ZentroMart Customer Support Agent | 22 October 2026

Participant/team: [Fill in before submission] Public repository: [Paste GitHub URL before submission]

1. Objective and approach

ZentroMart is a fictional Indian online store whose support team receives recurring questions about delivery, returns, refunds, cancellations and warranties. The objective is to build an assistant that grounds answers in company documents, handles follow-up turns, verifies callers before revealing order information, escalates cases when required, and supports voice input and output.

The solution uses Gemini for language generation and embeddings, a local retrieval index over the provided knowledge-base sections, synthetic CSV order records and supplied mock tools, and Gnani.ai for speech-to-text (STT) and text-to-speech (TTS). The application indexed 84 document sections and cached the index locally.

2. Implementation

Step 1: Prompt-only chatbot. A prompt-only customer-support assistant was implemented and tested. It can produce answers, but does not retrieve ZentroMart policies or query order records, so company-specific answers are less reliable.

Step 2: Retrieval-augmented generation (RAG). The assistant retrieves knowledge-base sections and cites their IDs. A grocery-return test retrieved RET-5, RET-2 and DWM-1. The answer correctly stated that groceries and fresh perishables are not returnable for a change of mind, while damaged, expired, wrong or missing items must be reported within 48 hours with photos. For an unsupported lifetime free screen-replacement question, the assistant did not confirm the benefit and instead cited current warranty rules.

Step 3: Conversation and mock tools. Selected manual tests covered phone-digit verification, order lookup, and escalation for a high-value damaged-TV claim. For a synthetic TV claim valued at ₹31,649, the agent identified Priority Resolution, declined to approve the replacement itself, retrieved DWM-3 and ESC-1/ESC-2, and reported ticket creation only after the mock tool confirmed success. This verifies selected scenarios, not all supplied conversations.

Step 4: Voice. The `gnani-vachana` SDK was installed and standalone Gnani STT/TTS calls passed. The batch runner processes audio through STT, the Step 3 agent and TTS, saving transcript, reply/evidence, JSON result and WAV reply. A generated audio question asking about the grocery return window was processed successfully; its response retrieved RET-5, RET-2 and DWM-1. The resulting WAV file existed and played correctly. Generated speech was used as input because the test laptop did not have a microphone.

3. Preliminary evaluation

Metric	Observed result	Interpretation
Question count	5 of 52	Initial smoke test only
Step 1 heuristic passes	2/5 (40%)	Not official course scoring
Step 2 heuristic passes	3/5 (60%)	Some calls affected by API failures
Step 2 required groups	4/4 checked groups	Heuristic check
Gold KB evidence	2/2 where defined	Small sample only

These are preliminary heuristic checks, not official capstone scores or a reliable estimate of unseen-test performance. The run encountered HTTP 503 service unavailability and HTTP 429 Gemini free-tier quota exhaustion. Affected calls did not produce normal answers. The 23 supplied multi-turn conversations have not been systematically evaluated.

4. Limitations and next improvements

The main operational limitation was Gemini quota and temporary service availability, which prevented completion of the full evaluation. The agent has been manually tested for selected grounding, verification and escalation scenarios, but the complete practice set has not been validated. The voice pipeline has succeeded for tested examples; more audio cases and a clean-environment reproduction test are still needed.

Next priorities are to evaluate additional practice cases when quota permits, systematically test multi-turn conversations, confirm the exact batch command from a clean setup, and inspect submission files for secret leakage. The project uses only synthetic course data. Gnani.ai is used for both STT and TTS as required. No claim is made that the prototype is production-ready or that all unseen test cases will pass.

5. Reproduction

Validate the course dataset: `python run.py check`

Start the RAG chatbot: `python run.py step2`

Start the conversation/tools agent: `python run.py step3`

Batch voice command: `python voice_batch.py --input-dir input_audio --output-dir voice_outputs`

For local execution, extract the course-provided dataset beside the source code as `Capstone_Project_Dataset_FTTT_2026/`. Do not include that dataset copy in the public repository or submission ZIP. Keep the real `.env` file local and include only the blank `.env.example` template.

Submission note: before submitting, replace the participant/team and repository placeholders on page 1. Confirm report length is at most four pages and include the public repository URL in README.md.